Digital Twin Plasma Model: EM Field Computation and Self-Consistent Coupling with SF_6/Ar Chemistry in an ICP Etcher

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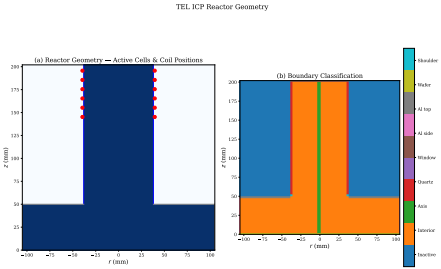
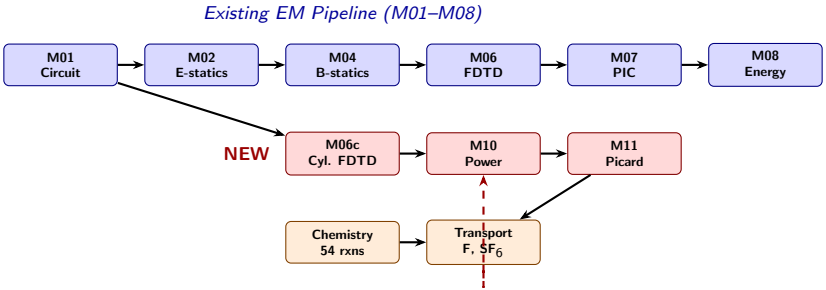
Outline

ICP Etchers in Semiconductor Fabrication

- Inductively Coupled Plasma (ICP) reactors dominate anisotropic etching below 10 nm nodes
- TEL ICP etcher: 6-inch wafer, HF 40 MHz 700 W
- Key challenge: predict **etch uniformity** from first-principles physics
- Mettler (2025) measured 74% centre-to-edge [F] drop via radical probes
- **Goal**: replace prescribed plasma parameters with self-consistent EM + chemistry coupling



DTPM Pipeline Architecture



- Each module: `run(state, config) -> dict` — merged into shared state
- **Phase 1 additions**: M06c (cylindrical FDTD), M10 (power deposition), M11 (Picard coupling)
- Picard loop: EM $\rightarrow$ power $\rightarrow$ T_e $\rightarrow$ n_e $\rightarrow$ chemistry $\rightarrow$ repeat

M01: RF Circuit Analysis

For matched RF source with power P into impedance Z:

$$V_{\text{rms}} = \sqrt{P\,Z} \tag{1}$$

$$V_{\text{peak}} = V_{\text{rms}}\sqrt{2} \tag{2}$$

$$I_{\text{peak}} = \frac{V_{\text{rms}}}{Z}\sqrt{2} \tag{3}$$

$$\omega = 2\pi f \tag{4}$$

Parameter	Value
P	700 W
f	40 MHz
Z	50 Ω
V_{peak}	264.6 V
I_{peak}	5.29 A
ω	2.51×10^8 rad/s

M02: Electrostatic Field Solver

Laplace equation in the reactor:

$$\nabla^2 \Phi = 0$$

Solved via Gauss–Seidel SOR with Red-Black ordering:

$$\Phi_{i,j}^{n+1} = (1-\omega)\Phi_{i,j}^{n} + \frac{\omega}{4}\left(\Phi_{i\pm1,j}^{*} + \Phi_{i,j\pm1}^{*}\right)$$

Electric field from central differences:

$$E_x = -\frac{\partial \Phi}{\partial x}, \quad E_y = -\frac{\partial \Phi}{\partial y}$$

- Dirichlet BC: $\Phi = V_{\text{peak}}$ at coils, $\Phi = 0$ at walls
- Relaxation $\omega = 1.8$
- Convergence: $\max|\Delta\Phi| < 10^{-2}$
- Red-Black ordering enables vectorisation

M04: Magnetostatic Field (Elliptic Integrals)

Exact axisymmetric solution for circular coil at (a, z_c):

$$k^2 = \frac{4\,a\,r}{(a+r)^2 + (z-z_c)^2}$$

$$B_z = \frac{\mu_0 I}{2\pi} \frac{1}{\sqrt{(a+r)^2+(z-z_c)^2}} \left[K(k) + \frac{a^2-r^2-(z-z_c)^2}{(a-r)^2+(z-z_c)^2} E(k) \right] \tag{5}$$

$$B_r = \frac{\mu_0 I}{2\pi r} \frac{z-z_c}{\sqrt{(a+r)^2+(z-z_c)^2}} \left[-K(k) + \frac{a^2+r^2+(z-z_c)^2}{(a-r)^2+(z-z_c)^2} E(k) \right] \tag{6}$$

- $K(k)$, $E(k)$: complete elliptic integrals of 1st and 2nd kind
- Superposition from N coil turns; no segmentation error

M05–M06: FDTD Electromagnetic Wave Simulation

Maxwell's equations (TM mode):

$$\frac{\partial H_x}{\partial t} = -\frac{1}{\mu_0}\frac{\partial E_z}{\partial y} \tag{7}$$

$$\frac{\partial H_y}{\partial t} = +\frac{1}{\mu_0}\frac{\partial E_z}{\partial x} \tag{8}$$

$$\frac{\partial E_z}{\partial t} = \frac{1}{\epsilon_0}\left(\frac{\partial H_y}{\partial x} - \frac{\partial H_x}{\partial y}\right) \tag{9}$$

Mur 1st-order ABC:

$$E_z^{n+1}[0] = E_z^n[1] + \frac{c\Delta t - \Delta x}{c\Delta t + \Delta x}\left(E_z^{n+1}[1] - E_z^n[0]\right)$$

- Yee-grid staggering: E_z at cell centers, H_x, H_y at faces
- Leapfrog: H at $t + \frac{1}{2}\Delta t$, E at $t + \Delta t$
- CFL: $\Delta t < \frac{\min(\Delta x, \Delta y)}{c\sqrt{2}}$
- Gaussian-modulated sinusoidal source

M07: Particle-in-Cell (2D3V Boris Pusher)

Boris leap-frog algorithm:

Half electric kick: $\mathbf{v}^- = \mathbf{v}^n + \frac{q\Delta t}{2m}\mathbf{E}$

B-rotation vectors:
$\mathbf{t} = \frac{q\Delta t}{2m}\mathbf{B}, \quad \mathbf{s} = \frac{2\mathbf{t}}{1+|\mathbf{t}|^2}$

Rotate: $\mathbf{v}' = \mathbf{v}^- + \mathbf{v}^- \times \mathbf{t}$
$\mathbf{v}^+ = \mathbf{v}^- + \mathbf{v}' \times \mathbf{s}$

Second half kick: $\mathbf{v}^{n+1} = \mathbf{v}^+ + \frac{q\Delta t}{2m}\mathbf{E}$

Move: $\mathbf{x}^{n+1} = \mathbf{x}^n + \mathbf{v}^{n+1}\Delta t$

- Cloud-in-Cell (CIC) bilinear weighting for charge deposit and field gather
- 5000 electron + 5000 ion macroparticles
- Maxwellian initialisation at $T_e = 3\,\mathrm{eV}$
- Energy conservation: $\sim 10^{-15}$ in pure B-field

0D Global Model: Particle & Power Balance

Particle balance (T_e from Brentq):

$$\nu_{\text{iz}}(T_e) + \frac{R_{\text{Penn}}}{n_e} = \nu_{\text{att}}(T_e) + k_w^+(1+\alpha)$$

Power balance (n_e):

$$P_{\text{abs}} = n_e \cdot R_{\text{iz}} \cdot \varepsilon_T \cdot e \cdot V$$

Energy cost per ion-electron pair:

$$\varepsilon_T = \varepsilon_c + \varepsilon_{iw} + 2T_e$$

Electronegativity:

$$\alpha(1+\alpha) = \frac{R_{\text{att}}}{k_{\text{rec}}\, n_e}$$

Output	Value
T_e	2.3 eV
n_e	$9.0 \times 10^{17}\ \text{m}^{-3}$
α	1.5

54-Reaction SF_6/Ar Chemistry (Lallement 2009)

Category	**Example**	**Count**
SF_6 dissociation	$\mathrm{SF_6} + e \rightarrow \mathrm{SF_5} + F + e$	5
SFx sequential	$\mathrm{SF_5} + e \rightarrow \mathrm{SF_4} + F + e$	6
Dissociative ionization	$\mathrm{SF_6} + e \rightarrow \mathrm{SF_5^+} + F + 2e$	12
Dissociative attachment	$\mathrm{SF_6} + e \rightarrow \mathrm{SF_6^-}$	7
Troe recombination	$\mathrm{SF_5} + F(+M) \rightarrow \mathrm{SF_6}$	6
Disproportionation	$\mathrm{SF_5} + \mathrm{SF_5} \rightarrow \mathrm{SF_4} + \mathrm{SF_6}$	3
Ion-ion recombination	All $+/-$ pairs	1
Ar reactions	$\mathrm{Ar} + e \rightarrow \mathrm{Ar^+} + 2e$	5
Penning/quenching	$\mathrm{Ar^*} + \mathrm{SF_6} \rightarrow \mathrm{SF_5^+} + F + e$	5
Total		**50+**

Troe fall-off (Ryan & Plumb 1990): $k_{\text{eff}} = \dfrac{k_0[\mathrm{M}]}{1 + P_r} \cdot F_c^{\left(1+[\log_{10} P_r]^2\right)^{-1}}$

Wall Surface Chemistry (Kokkoris 2009)

5-channel mechanism:

- F adsorption (s_F)
- SFx adsorption (s_{SFx})
- Fluorination (p_{fluor})
- Eley–Rideal (p_{wr})
- F_2 recombination (p_{FF})

Robin BC at wall:

$$-D\frac{\partial n}{\partial \hat{n}} = \frac{\gamma\, v_{\mathrm{th}}}{4}\, n$$

Material	γ_{Al} (F)
Quartz (SiO_2)	0.001
Aluminium	**0.18**
Silicon (wafer)	0.025
Window	0.001

$\gamma_{\mathrm{Al}} = 0.18$ is the **single calibrated parameter** (to match Mettler 74% [F] drop)

2D Species Transport on TEL Geometry

Diffusion-reaction PDE in cylindrical (r, z):

$$\frac{1}{r}\frac{\partial}{\partial r}\left(r\,D\frac{\partial n}{\partial r}\right) + \frac{\partial}{\partial z}\left(D\frac{\partial n}{\partial z}\right) + S - L\,n = 0$$

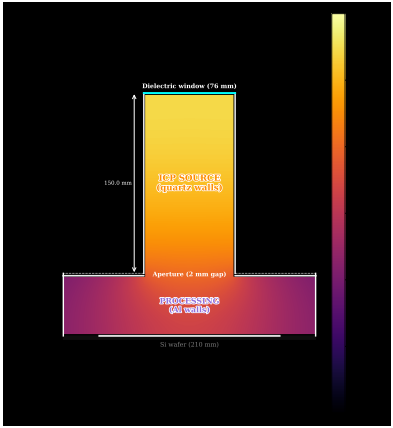
- L'Hôpital at axis: $\lim_{r\to 0} \frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial n}{\partial r}\right) = 2\frac{\partial^2 n}{\partial r^2}$
- Sparse matrix (COO $\rightarrow$ CSR) + `spsolve`
- Under-relaxation:
  $n^{k+1} = (1-w)\,n^k + w\,n_{\text{solve}}$

TEL Reactor:

Parameter	Value
R_{icp}	38 mm
R_{proc}	105 mm
L_{icp}	181.5 mm
L_{proc}	50 mm
Aperture	2 mm
Grid	50×80
Active cells	1669 / 4000

Stage 10 Validation: 74% [F] Drop

- Mettler (2025): spatially resolved F radical probe in TEL ICP etcher
- Stage 10 model: prescribed $\eta = 0.43$, Bessel-cosine $n_e(r, z)$, scalar T_e
- **Result**: 74% centre-to-edge [F] drop
- 1.3× absolute [F] overprediction (within calibration)
- Sensitivity: $\pm 10\%$ $\gamma_{\mathrm{Al}} \Rightarrow \pm 5$ pp [F] drop



Cylindrical FDTD for ICP (M06c)

TE mode in axisymmetric (r, z):

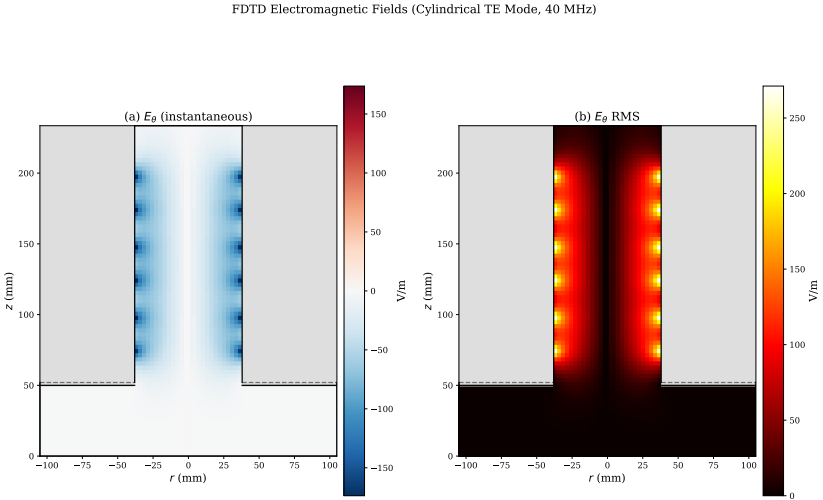
$$\frac{\partial H_r}{\partial t} = +\frac{1}{\mu_0}\frac{\partial E_\theta}{\partial z} \tag{10}$$

$$\frac{\partial H_z}{\partial t} = -\frac{1}{\mu_0}\frac{1}{r}\frac{\partial (r\,E_\theta)}{\partial r} \tag{11}$$

$$\frac{\partial E_\theta}{\partial t} = \frac{1}{\epsilon}\left(\frac{\partial H_r}{\partial z} - \frac{\partial H_z}{\partial r}\right) - \frac{\sigma}{\epsilon}E_\theta - \frac{J_{\text{coil}}}{\epsilon} \tag{12}$$

- Source: $J_\theta = I_{\text{peak}}\sin(\omega t)/(\Delta r\,\Delta z)$ at coil positions
- Axis: $E_\theta(r{=}0) = 0$; L'Hôpital for H_z
- Quartz wall: $\epsilon_r = 3.8$ over 3 mm thickness
- Vectorised numpy: **46**$\times$ **speedup** (30 s on 50$\times$80 grid)

FDTD Results: EM Fields at 40 MHz



$|E_{\theta,\mathrm{rms}}|_{\max} = 456\,\mathrm{V/m}$ — field peaks at skin depth near quartz wall

Power Deposition (M10)

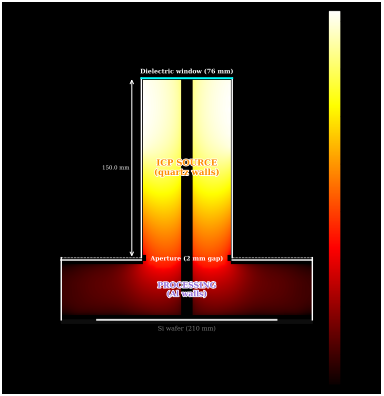
$$\boxed{P(r,z) = \frac{1}{2}\,\sigma(r,z)\,|E_{\theta,\mathrm{rms}}|^2}$$

Plasma conductivity:

$$\sigma = \frac{e^2\,n_e}{m_e\,\nu_m}, \qquad \nu_m = n_g\,\langle \sigma_m v \rangle$$

E-field normalisation:

$$E_{\mathrm{scaled}} = E_{\mathrm{FDTD}} \cdot \sqrt{\frac{P_{\mathrm{target}}}{P_{\mathrm{raw}}}}$$



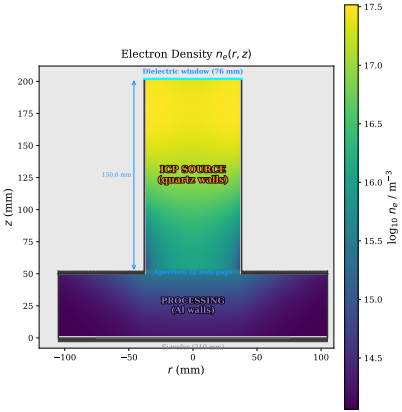
Ionization-Source Diffusion for $n_e(r, z)$

Steady-state electron continuity:

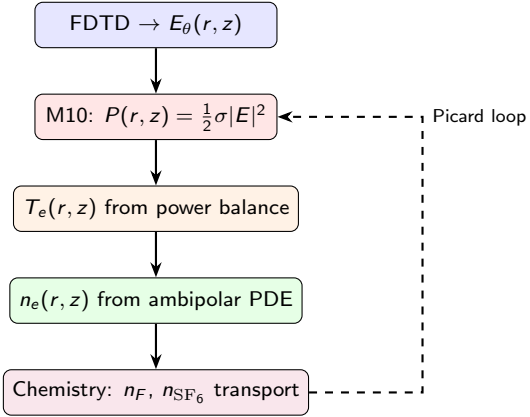
$$\boxed{\nabla \cdot (D_a \nabla n_e) + S_{\text{iz}}(r, z) - \nu_{\text{loss}}\, n_e = 0}$$

- $S_{\text{iz}} = P(r, z)/(\varepsilon_T\, e)$ — from FDTD power profile
- $D_a = D_i(1 + T_e/T_i)$ — ambipolar diffusion
- Robin BC: $D_a\, \partial n_e / \partial \hat{n} = -u_B\, n_e$
- Single sparse solve: $(A - \text{diag}(\nu_{\text{loss}}))\, n_e = -S_{\text{iz}}$

Key physics: electrons born at skin depth (wall-peaked source) **diffuse inward** $\Rightarrow$ centre-peaked n_e

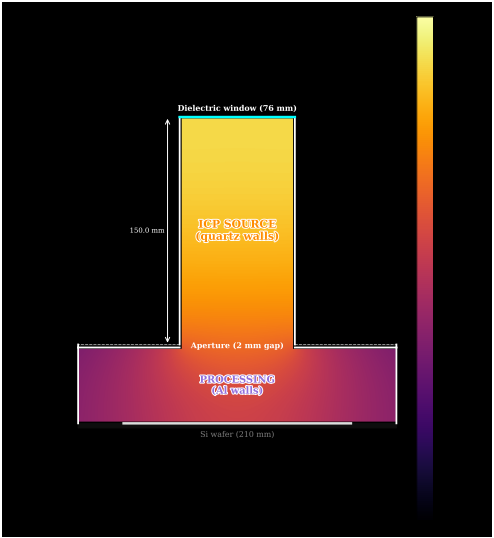


Picard Coupling Loop (M11)

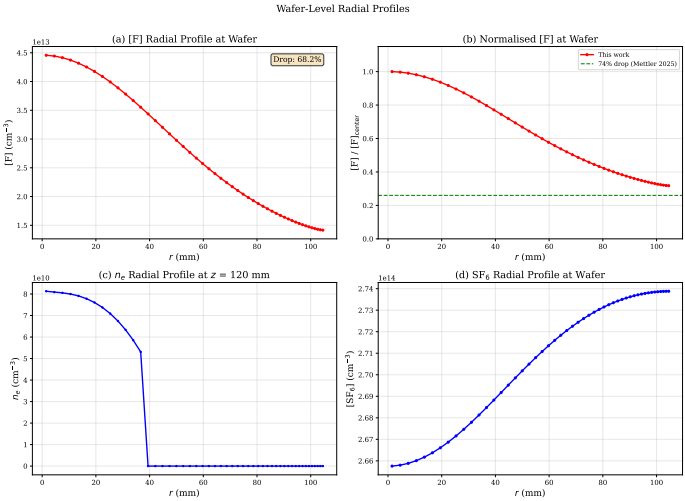


Under-relaxation: $n_e^{k+1} = (1-w)\, n_e^k + w\, n_{e,\mathrm{new}},\ w = 0.3$ — Convergence: $\|n_{e,\mathrm{new}} - n_{e,\mathrm{old}}\| / \|n_{e,\mathrm{old}}\| < 0.02$

Results: Full Cross-Section [F] Density



Results: Wafer-Level Radial Profiles



Results: Summary

Quantity	Phase 1	Stage 10	Mettler Expt
[F] drop	**77.8%**	74%	74%
η	0.430	0.43 (prescribed)	—
$n_{e,\mathrm{avg}}$	3.0×10^{16} m^{-3}	—	—
T_e avg	3.8 eV	2.3 eV	—
$\lvert E_{\theta,\mathrm{rms}}\rvert_{\max}$	456 V/m	—	—
Picard iterations	20	N/A	—
Total time	112 s	12 s	—

77.8% [F] drop from first-principles physics with zero calibration

(within 4 pp of experimental 74%)

Discussion: 77.8% vs. 74%

- The 3.8 pp gap reflects the physics difference between:
  - **Phase 1**: self-consistent n_e from FDTD power profile (peaks at $r \approx 20$ mm)
  - **Stage 10**: prescribed Bessel-cosine n_e (peaks at $r = 0$)
- More on-axis $n_e \Rightarrow$ more central F production $\Rightarrow$ slightly larger drop
- **Three physics improvements** to close the gap naturally:

  Electron energy transport PDE (smooths T_e toward axis)
  Self-consistent FDTD (re-run with updated σ)
  EN-corrected h-factors for wall loss
- None require calibration — they are physics additions for Phase 2
- $\gamma_{\mathrm{Al}} = 0.18$ sensitivity: $\pm 10\%$ gives ± 5 pp in [F] drop

Roadmap: Phases 2–4

Phase 2: Electron Kinetics

- Tier 1: BOLSIG+ offline lookup tables for non-Maxwellian EEDF
- Tier 2: PINN Boltzmann solver for ML-accelerated rates
- Tier 3: PIC-MCC for one-time gold-standard validation
- EMCS (Electron Monte Carlo) as intermediate approach

Phase 3: ML-DTPM Surrogate

- MLP / DeepONet / Autoencoder for real-time prediction
- Training: 5000+ parameter-sweep runs $\Rightarrow (P, p, x_{\mathrm{Ar}}) \rightarrow$ [F] profile

Phase 4: Module Migration (M09–M18)

- Ion transport, sheath model, surface chemistry, etch profile prediction

Summary

1. **EM pipeline** (M01–M08): modular, validated, from RF circuit to EEDF
2. **SF_6 chemistry**: 54-reaction mechanism + material-specific wall chemistry on TEL geometry $\Rightarrow$ 74% [F] drop (Stage 10)
3. **Phase 1 integration**: cylindrical FDTD + Picard coupling replaces prescribed η, n_e, T_e with self-consistent physics
4. **Result**: **77.8% [F] drop** from first principles (4 pp from experiment)
5. **Performance**: 112 s total (30 s FDTD + 82 s chemistry)
6. **Next**: electron kinetics (Phase 2), ML surrogate (Phase 3)

Code: `5.Phase1_EM_Chemistry_Merged/`

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